

BOM Manual Downloader

How the App Searches the Web and Verifies Every Downloaded Document

Technical explanation of the search cascade, candidate ranking,
and the content-verification criteria that decide whether a PDF
is accepted as a genuine manual or datasheet.

July 2026 | reflects bom_downloader.py as shipped

1. The Pipeline at a Glance

For every part the engine runs the same five stages. A part leaves the pipeline at the first stage that produces a verified document.

Stage	What happens
1. Catalog lookup	Confident hit in your own datasheet library (exact part number in a filename, inside a PDF, or pinned in catalog_map.txt) — the web is skipped entirely
2. Search cascade	Up to 7 tiers of web sources, most trustworthy first, under a 75-second per-part time budget
3. Ranking	Every candidate URL is scored; best candidates are tried first
4. Download + verify	Each candidate PDF is downloaded and its CONTENT is classified; wrong or junk documents are rejected and the next candidate is tried (max 8 downloads per part)
5. Retry pass	Parts that found nothing are re-run one-at-a-time at the end with search engines forced on

2. The Web Search Cascade

The guiding principle: the manufacturer's own website is always more accurate than any search engine, and search engines are always more accurate than generic scraping. The engine therefore works down this ladder and stops as soon as trustworthy documents are in hand.

Tier 1 — DirectProbe (manufacturer CDN patterns)

For known manufacturers the app constructs the datasheet URL directly from hard-coded CDN patterns (e.g. Rittal's pdf-creator service, DigiKey media, Rockwell literature) and confirms each with a quick HEAD request. Deterministic, instant, and by construction part-exact.

Tier 1.5 — Manufacturer portal scraping

The app scrapes the manufacturer's own product/documentation page for the part. If this (or Tier 1) yields usable official documents, the search-engine cascade is skipped entirely. If the official docs contain only datasheets, exactly ONE extra search query is spent looking for a matching manual, then the part moves on.

Tier 2 — Component databases (optional API keys)

Nexar/Octopart and similar part databases resolve manufacturer + part number to official datasheet links. Only active when an API key is configured.

Tier 3 — Site-restricted search

For manufacturers with a known documentation domain the app issues site:-restricted DuckDuckGo queries (e.g. site:literature.rockwellautomation.com "1756-EN2T") — this returns exact documents instead of random distributor pages. Result pages that are not PDFs are scraped for PDF links in parallel.

Tier 4 — Distributor scrapers

Mouser and RS Components product pages are scraped for their datasheet links — no API key needed. Distributors host verified copies of manufacturer documents.

Tier 5 — Broad web search

General queries built from the part data — manufacturer + quoted part number + description keywords + the words 'manual' and 'datasheet', including a dotless part-number variant (8108.245 → 8108245). Non-PDF result pages are scraped for PDF links. If candidates already exist this tier only tops up with 2 queries instead of ~7.

Tier 6 — Last-ditch PDF search

Only when still empty-handed: a plain '<part> pdf' query.

Tier 7 — Family-level fallback

Some configured articles only have family-level literature. These hits are clearly flagged '[family doc]' and are never auto-picked as part-specific documents.

Guard rails around the cascade

- 75-second hard budget per part: whenever it expires, whatever was found so far is ranked and the download stage starts.
- Circuit breaker: any backend that keeps failing (bad key, bot-block) is disabled for the rest of the run after 3 failures.
- DuckDuckGo rate-limit protection: queries are serialized through a global queue; if DDG rate-limits, it is paused globally and parts that depend on it fail fast instead of queueing for minutes.
- Relevance filter before anything is accepted as a candidate: the part number must appear as a WHOLE token in the URL/title/snippet (EN2T never matches EN2TSC), or the link must be on the manufacturer's own domain and contain a significant chunk of the part number.
- End-of-run retry: every part that found nothing is re-searched one-at-a-time with engines forced — the same thing the manual 'Search again' button does.

3. Candidate Ranking — Which Link Gets Tried First

Every candidate URL receives a score; downloads happen in score order. The main factors and their weights:

Signal	Points
Part number matches as a whole token in URL / title / snippet	+20
Manufacturer's OWN website hosts the file	+25
Known manufacturer CDN domain	+10
Known distributor domain (Mouser, RS, Farnell...)	+5
Filename/title says it is a MANUAL	+14
Filename/title says it is a datasheet	+6

Filename/title says license / certificate / CAD drawing / brochure	−40
English language code in filename (..._EN.pdf)	+6
Foreign-language-only filename (..._DE.pdf)	−10
Each description keyword found (up to 4)	+3 each
Recent year in URL/title	up to +4
Old year (5–15 years back)	−4
Unknown domain, not manufacturer or distributor	−5

Two practical consequences: an official manufacturer manual in English virtually always outranks everything else, and anything whose NAME already says 'certificate' or 'CAD drawing' is effectively never downloaded at all.

4. Download Policy — Two Phases

Phase 1: candidates are downloaded in rank order until the configured number of documents per part is accepted (default 1) or a real manual is found — capped at 8 download attempts per part.

Phase 2 — the manual hunt: if Phase 1 only produced datasheets, the engine spends at most 2 MORE downloads, and only on candidates whose filename explicitly says 'manual'. A verified manual then replaces a lower-quality datasheet.

- Encrypted PDFs: files that are merely permission-protected (empty password) are silently rewritten without encryption; truly password-locked files are rejected.
- Every accepted file is renamed to a stable scheme: NNN_MFR_PART__Title.pdf where NNN is the BOM position.

5. Content Verification — Is This Actually the Right Manual?

This is the decisive quality gate (VERIFY_CONTENT, on by default): every downloaded PDF is opened and CLASSIFIED BY WHAT IS INSIDE IT, not by what the link promised. A file is only kept if it passes all three checks below.

Check 1 — Is it a readable document?

- Text is extracted from a sample of pages (pdfplumber, falling back to pypdf).
- Unparseable files are rejected — unless they are structurally complete PDFs served from the manufacturer's OWN domain with a manual/datasheet filename, which are trusted on source.
- Image-only scans (old paper manuals) are accepted only when the filename says manual/datasheet or the page count (≥5) makes a drawing/certificate unlikely.

Check 2 — Is it about THIS part?

- Exact match: the full part number appears as a whole token in the extracted text — strongest confirmation.
- Fragment match: significant chunks of the part number appear — weaker, accepted only with other evidence.
- Source-link match: manufacturer manuals are often image-heavy and never spell the part number as text, but the official download URL literally contains it (.../8108245_Instructions_EN). An exact whole-token match in the source URL counts as confirmation.
- Strict rule for all-numeric part numbers (e.g. Rittal 8601020): a fragment match is NOT enough, because 8601000 vs 8601020 are different variants — the exact number must appear in the text or the source link, otherwise the file is rejected as 'wrong part'.
- A document that never mentions the part in any of these ways is rejected outright.

Check 3 — What KIND of document is it?

The extracted text is scored against six weighted keyword profiles; the highest-scoring profile (minimum score 3) classifies the document:

Class	Typical scoring phrases	Outcome
Manual	table of contents, user/instruction/installation/operating manual, user guide, original instructions	accepted — highest quality
Datasheet	datasheet, technical data/specifications, ordering information, electrical characteristics, maximum ratings	accepted
License	license agreement, EULA, licensor/licensee, GPL	rejected
Certificate	declaration of conformity, EU/EC declaration, notified body, certificate number	rejected
CAD drawing	do not scale, drawing number, third-angle projection, dimensions in mm, sheet 1 of	rejected
Brochure	brochure, your benefits, product overview, highlights	rejected

Structural evidence is added on top of the keywords:

- 8 or more pages → manual score +2 (licenses, certificates and drawings are rarely that long).
- 3 pages or fewer with sparse text density → CAD-drawing score +3 (typical title-block layout).
- The filename's own claim (manual/datasheet) adds +2 to that class — a hint, never a verdict.

Quality score — picking the best file

Files that survive all checks get a quality score used to keep the best document per part: manual = 5, datasheet = 3, accepted scan = 2, unknown = 1; bonuses for an exact part-number match

(+1), manuals of 10+ pages (+1), and 2+ BOM-description keywords found in the text (+1). This is why a verified 40-page installation manual always wins over a 2-page product flyer.

What you see in the log

Every verification decision is logged in one line, e.g.:

Verified: DATASHEET [6p, signals={'datasheet': 12}, part-match=exact, desc-match=2 keyword(s)]

...and every rejection states its reason ('part number not found anywhere in document', 'looks like a different variant', 'filename indicates non-manual', ...), so a human can always audit why a file was kept or discarded.

6. Summary

A document ends up in your record book only if: (1) it came from the most trustworthy source available for that part, (2) its link survived whole-token part-number relevance filtering, (3) it won the candidate ranking, and (4) its actual content identified it as a manual or datasheet that mentions this exact part. Anything else — certificates, CAD drawings, licenses, brochures, wrong variants, unreadable files — is rejected with a logged reason, and the engine moves to the next candidate.